```
!pip install cmdstanpy

from cmdstanpy import install_cmdstan
install_cmdstan()
```

```
Requirement already satisfied: cmdstanpy in /usr/local/lib/python3.12/dist-packages (1.3.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from cmdstanpy) (2.2.2)
Requirement already satisfied: numpy>=1.21 in /usr/local/lib/python3.12/dist-packages (from cmdstanpy) (2.0.2)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from cmdstanpy) (4.67.3)
Requirement already satisfied: stanio<2.0.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from cmdstanpy) (0.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->cmdstanpy) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->cmdstanpy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->cmdstanpy) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->cmdstanpy) (1.17.0)
CmdStan install directory: /root/.cmdstan
CmdStan version 2.38.0 already installed
Test model compilation
True
```

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

from sklearn.model_selection import train_test_split
from cmdstanpy import CmdStanModel
```

```
from google.colab import drive
drive.mount('/content/drive')

cali_house = pd.read_csv('/content/drive/Shareddrives/5440gaussian process/5440gaussian process/california housing.csv')
cali_house.head()
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

About the Dataset

↳ 已隐藏 10 个单元格

Gaussian Process Regression Formulas

Model Specification:

Prior over functions

We place a Gaussian Process prior on the latent function:

$$f(x) \sim \mathcal{GP}(0, k(x, x'))$$

Observation model

Observations are modeled as:

$$y = f(x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Kernel function (RBF)

We use the radial basis function (RBF) kernel:

$$k(x, x') = \alpha^2 \exp\left(-\frac{\|x - x'\|^2}{2\rho^2}\right)$$

Joint Distribution

Let:

- X denote training inputs
- X_* denote test inputs

Then the joint distribution is:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K(X,X) + \sigma^2 I & K(X,X_*) \\ K(X_*,X) & K(X_*,X_*) \end{bmatrix}\right)$$

Covariance Matrices

We define:

$$K = K(X,X) + \sigma^2 I$$
$$K_s = K(X,X_*)$$
$$K_{ss} = K(X_*,X_*)$$

Posterior Predictive Distribution

Conditioning on observed data, the predictive distribution is:

$$f_* \mid y \sim \mathcal{N}(\mu_*, \Sigma_*)$$

Predictive mean

$$\mu_* = K_s^T K^{-1} y$$

Predictive covariance

$$\Sigma_* = K_{ss} - K_s^T K^{-1} K_s$$

Implementation Formulation

Instead of explicitly computing K^{-1}, we solve:

$$\alpha = K^{-1} y$$
$$\mu_* = K_s^T \alpha$$

Posterior Predictive for Observations

Including observation noise:

$$y_* \sim \mathcal{N}(\mu_*, \Sigma_* + \sigma^2 I)$$

Posterior Predictive Sampling

We draw samples:

$$y_*^{(s)} \sim \mathcal{N}(\mu_*, \Sigma_* + \sigma^2 I)$$

Numerical Stabilization (Jitter)

To ensure numerical stability, we modify:

$$K \leftarrow K + \epsilon I$$

where ϵ is a small constant (e.g., 10^{-8}).

Summary

Gaussian Process regression models the joint distribution of training and test outputs as a multivariate normal distribution. Predictions are obtained by conditioning this joint distribution, yielding closed-form expressions for both the predictive mean and uncertainty.

```
stan_data = {
        'N_train': X_train_scaled.shape[0],
        'N_test': X_test_scaled.shape[0],
        'D': X_train_scaled.shape[1],
        'X_train': X_train_scaled,
        'X_test': X_test_scaled,
        'y_train': y_train_scaled
}
```

```
model = CmdStanModel(stan_file='/content/drive/Shareddrives/5440gaussian_process/5440gaussian_process/gp_california.stan')

fit = model.sample(
        data=stan_data,
        chains=4,
        parallel_chains=4,
        iter_warmup=1000,
        iter_sampling=1000,
        seed=42
)
```

```
chain 1: 100%                                    2000/2000 [07:24<00:00,  4.04it/s, (Sampling completed)]
chain 2: 100%                                    2000/2000 [07:24<00:00,  4.26it/s, (Sampling completed)]
chain 3: 100%                                    2000/2000 [07:24<00:00,  4.37it/s, (Sampling completed)]
chain 4: 100%                                    2000/2000 [07:24<00:00,  4.09it/s, (Sampling completed)]

Exception: cholesky_decompose: A is not symmetric. A[1,2] = -nan, but A[2,1] = -nan (in 'gp_california.stan', line 40, column 2 to column 30)
   Exception: cholesky_decompose: Matrix m is not positive definite (in 'gp_california.stan', line 40, column 2 to column 30)
Exception: cholesky_decompose: A is not symmetric. A[1,2] = -nan, but A[2,1] = -nan (in 'gp_california.stan', line 40, column 2 to column 30)
Consider re-running with show_console=True if the above output is unclear!
```

Diagnostics

```
summary_df = fit.summary()
print(summary_df.columns)
```

```
Index(['Mean', 'MCSE', 'StdDev', 'MAD', '5%', '50%', '95%', 'ESS_bulk',
       'ESS_tail', 'ESS_bulk/s', 'R_hat'],
      dtype='object')
```

```
params = ['alpha', 'rho', 'sigma']
print(summary_df.loc[params, ['Mean', 'StdDev', 'R_hat', 'ESS_bulk', 'ESS_tail']])
```

```
           Mean    StdDev    R_hat  ESS_bulk  ESS_tail
alpha  0.918281  0.233114  1.00542  1158.900   663.511
rho    0.389081  0.290146  1.00602   750.555   801.452
sigma  0.772572  0.059097  1.00153  1049.910   922.148
```

```
sampler_diagnostics = fit.diagnose()
print(sampler_diagnostics)
```

```
Checking sampler transitions treedepth.
Treedepth satisfactory for all transitions.

Checking sampler transitions for divergences.
No divergent transitions found.

Checking E-BFMI - sampler transitions HMC potential energy.
E-BFMI satisfactory.

Rank-normalized split effective sample size satisfactory for all parameters.

Rank-normalized split R-hat values satisfactory for all parameters.

Processing complete, no problems detected.
```

now that seems pretty healthy - converged!

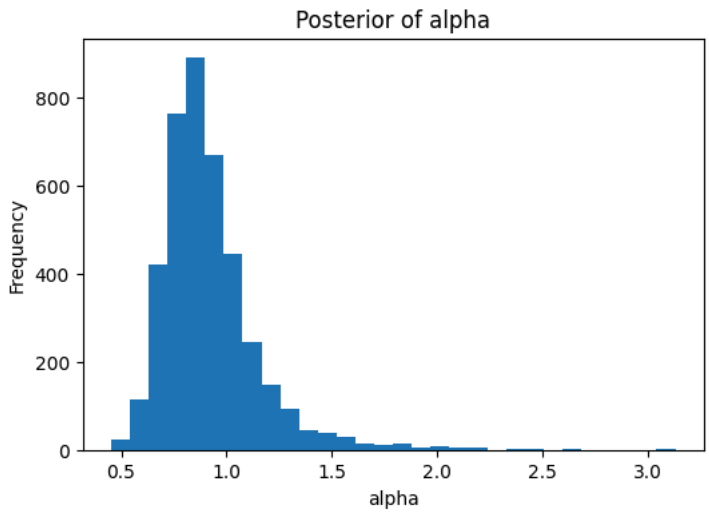
Posterior summaries

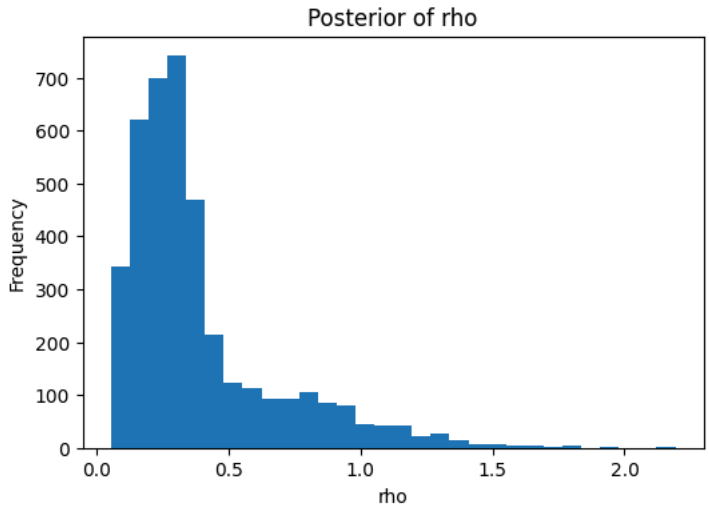
```
posterior = fit.draws_pd()

params_of_interest = ['alpha', 'rho', 'sigma']
posterior_summary = summary_df.loc[
        params_of_interest,
        ['Mean', 'StdDev', '5%', '50%', '95%', 'R_hat']
]
print(posterior_summary)
```

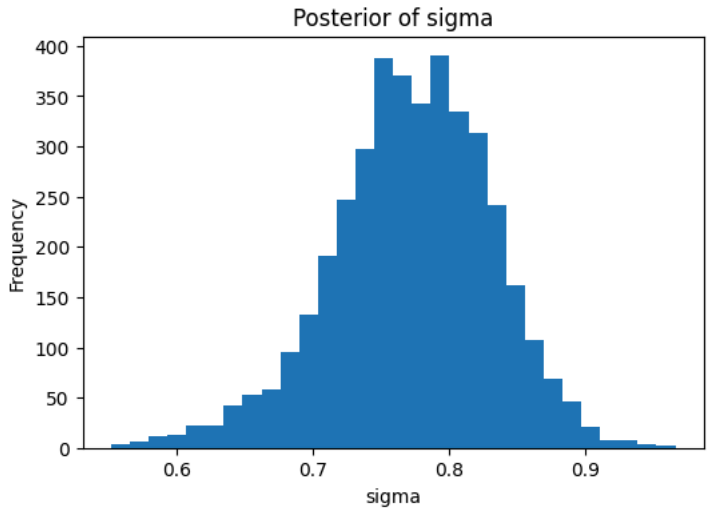
```
          Mean    StdDev        5%       50%      95%    R_hat
alpha  0.918281  0.233114  0.651976  0.873101  1.32777  1.00542
rho    0.389081  0.290146  0.096951  0.299800  1.01812  1.00602
sigma  0.772572  0.059097  0.669029  0.774810  0.86445  1.00153
```

Visualize Posterior

```
for param in params_of_interest:
    plt.figure(figsize=(6, 4))
    plt.hist(posterior[param], bins=30)
    plt.title(f'Posterior of {param}')
    plt.xlabel(param)
    plt.ylabel('Frequency')
    plt.show()
```







Posterior predictive checks

```
# Extract posterior predictive draws for test points
y_test_rep = fit.stan_variable('y_test_rep')      # shape: (draws, N_test)

# Posterior predictive mean on standardized scale
y_test_pred_mean_scaled = y_test_rep.mean(axis=0)

# Transform back to original scale
y_test_pred_mean = y_test_pred_mean_scaled * y_std + y_mean
```

Posterior Predictive Performance

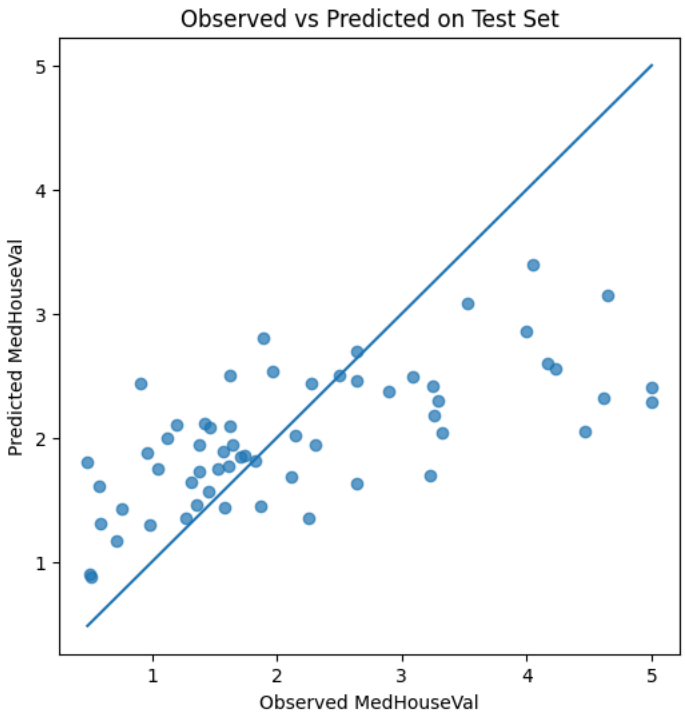
predicted vs observed

The scatter plot of observed versus predicted values shows a positive relationship, indicating that the model captures some of the underlying structure in the data.

However, the predictions tend to be biased toward the mean: lower observed values are overestimated, while higher observed values are underestimated. This pattern suggests that the model is unable to fully capture the variability in housing prices.

This behavior is expected, as the model only uses spatial features (latitude and longitude) and does not incorporate other important predictors such as income, housing characteristics, or local economic factors.

```
plt.figure(figsize=(6, 6))

plt.scatter(y_test, y_test_pred_mean, alpha=0.7)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()])

plt.xlabel('Observed MedHouseVal')
plt.ylabel('Predicted MedHouseVal')
plt.title('Observed vs Predicted on Test Set')
plt.show()
```

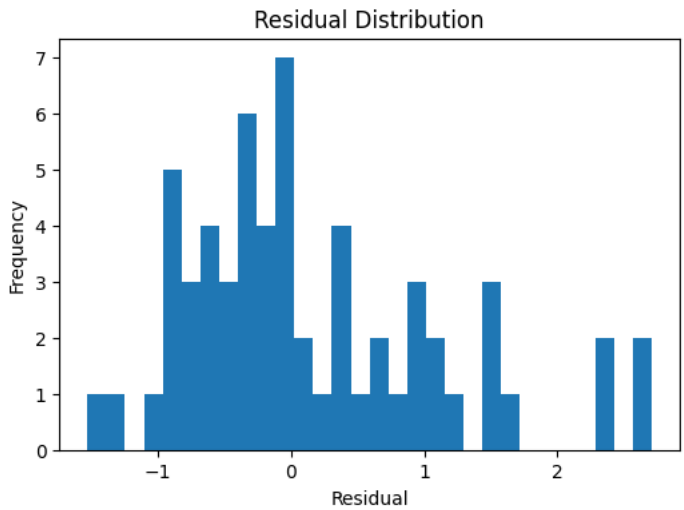


Residual Analysis

The residuals are generally centered around zero, indicating that the model is approximately unbiased on average. However, the distribution exhibits a noticeable right tail, suggesting that the model tends to underestimate higher housing prices.

This pattern is consistent with the observed vs. predicted plot, where extreme values are shrunk toward the mean. This behavior reflects the limitations of the model, which relies only on spatial features and cannot fully capture all sources of variability in housing prices.

```
residuals = y_test - y_test_pred_mean

plt.figure(figsize=(6, 4))

plt.hist(residuals, bins=30)

plt.xlabel('Residual')
plt.ylabel('Frequency')
plt.title('Residual Distribution')
plt.show()
```



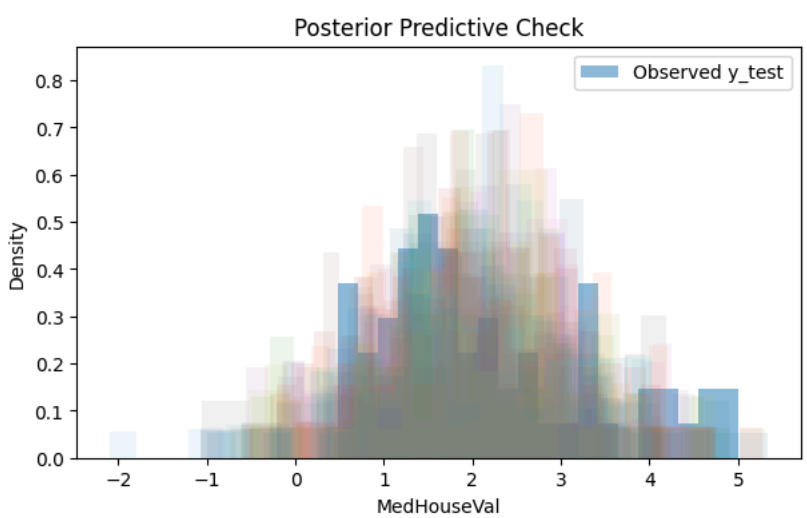
Posterior Predictive Check

The predictive distributions broadly align with the observed data in the central region, indicating that the model captures the overall structure of the housing price distribution.

However, discrepancies are evident in the tails, particularly for higher values. The model tends to underestimate extreme housing prices, producing smoother distributions with less mass in the upper range.

This suggests that while the Gaussian Process captures large-scale spatial trends, it lacks the flexibility to model sharp variations and extreme values. This limitation is expected given that only spatial features (latitude and longitude) are used in the model.

```
# a few posterior predictive draws and transform back
num_draws_to_plot = 30
draw_indices = np.random.choice(y_test_rep.shape[0], size=num_draws_to_plot, replace=False)

plt.figure(figsize=(7, 4))
plt.hist(y_test, bins=20, alpha=0.5, density=True, label='Observed y_test')

for idx in draw_indices:
    y_rep_original = y_test_rep[idx] * y_std + y_mean
    plt.hist(y_rep_original, bins=20, alpha=0.08, density=True)

plt.xlabel('MedHouseVal')
plt.ylabel('Density')
plt.title('Posterior Predictive Check')
plt.legend()
plt.show()
```



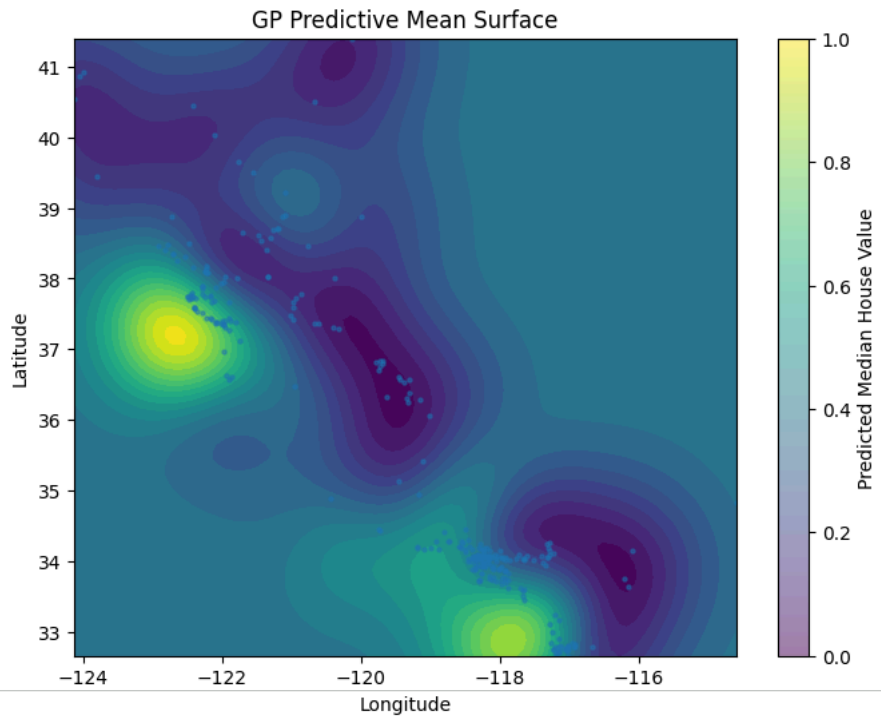
Heatmap

```
# create a grid
lon_min, lon_max = df['Longitude'].min(), df['Longitude'].max()
lat_min, lat_max = df['Latitude'].min(), df['Latitude'].max()

lon_grid = np.linspace(lon_min, lon_max, 80)
lat_grid = np.linspace(lat_min, lat_max, 80)

lon_mesh, lat_mesh = np.meshgrid(lon_grid, lat_grid)

X_grid = np.column_stack([lat_mesh.ravel(), lon_mesh.ravel()])
```

```
# standardize
X_grid_scaled = x_scaler.transform(X_grid)
```

```
# get the values

# Posterior mean hyperparameters
alpha = 0.918281
rho = 0.389081
sigma = 0.772572
jitter = 1e-5

# Training data in standardized scale
Xtr = X_train_scaled
ytr = y_train_scaled

# Grid data in standardized scale
Xte = X_grid_scaled

def rbf_kernel(X1, X2, alpha, rho):
        # Squared Euclidean distance matrix
        sqdist = (
                np.sum(X1**2, axis=1)[:, None]
                + np.sum(X2**2, axis=1)[None, :]
                - 2 * X1 @ X2.T
        )
        return (alpha**2) * np.exp(-0.5 * sqdist / (rho**2))

# Covariance blocks
K = rbf_kernel(Xtr, Xtr, alpha, rho)
K += (sigma**2 + jitter) * np.eye(len(Xtr))

K_s = rbf_kernel(Xtr, Xte, alpha, rho)
K_ss = rbf_kernel(Xte, Xte, alpha, rho)
K_ss += jitter * np.eye(len(Xte))

# Solve for predictive mean
K_inv_y = np.linalg.solve(K, ytr)
f_mean_scaled = K_s.T @ K_inv_y

# Predictive covariance
K_inv_Ks = np.linalg.solve(K, K_s)
f_cov_scaled = K_ss - K_s.T @ K_inv_Ks

# Convert back to original scale
f_mean = f_mean_scaled * y_std + y_mean
f_sd = np.sqrt(np.clip(np.diag(f_cov_scaled), 0, None)) * y_std

mean_map = f_mean.reshape(lat_mesh.shape)
sd_map = f_sd.reshape(lat_mesh.shape)
```

```
# Predictive Mean Surface heatmap
plt.figure(figsize=(8, 6))

plt.contourf(lon_mesh, lat_mesh, mean_map, levels=30)
plt.scatter(df['Longitude'], df['Latitude'], s=5, alpha=0.5)

plt.colorbar(label='Predicted Median House Value')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.title('GP Predictive Mean Surface')
plt.show()
```



The GP predictive mean surface reveals a smooth, non-linear spatial structure in housing prices across California. High-value regions are concentrated around major urban areas, such as the Bay Area and Southern California, indicating strong spatial dependence.

The smooth transitions between regions reflect the properties of the RBF kernel, which enforces local continuity in the latent function.

```
# Uncertainty heatmap
plt.figure(figsize=(8, 6))

plt.contourf(lon_mesh, lat_mesh, sd_map, levels=30)
plt.scatter(df['Longitude'], df['Latitude'], s=5, alpha=0.5)
```